

Aharonov-Bohm Effect

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The Schroedinger equation in magnetic field is

$$\begin{aligned} & \left(-i\nabla - q\vec{A} \right)^2 \psi = 2im\partial_t \psi \\ \iff & \left(-\nabla^2 + q^2 A^2 + 2iq\vec{A} \cdot \nabla + iq(\nabla \cdot \vec{A}) \right) \psi = 2mE\psi \end{aligned} \quad (1.1)$$

where $\partial_t \psi = 0$ and $\hbar = 1$. We consider the domain in which $B \equiv \nabla \times A = 0$, and let $\psi \equiv \varphi e^{iq \int^{\vec{x}} \vec{A} \cdot d\vec{x}} \equiv \varphi e^{iS(\vec{x})}$. Then

$$\begin{aligned} \nabla \psi &= e^{iS} \left(\nabla \varphi + iq\vec{A}\varphi \right) \\ \nabla^2 \psi &= e^{iS} \left(\nabla^2 + 2iq\vec{A} \cdot \nabla + iq\nabla \cdot \vec{A} - q^2 A^2 \right) \varphi \end{aligned} \quad (1.2)$$

$$\Rightarrow -\nabla^2 \varphi = 2im\partial_t \varphi \quad (1.3)$$

Its time independent form is

$$\nabla^2 \varphi + k^2 \varphi = 0 \quad (1.4)$$

where $k^2 = 2mE$. This is a Helmholtz equation, whose solution is

$$\varphi(r, \theta) = \sum_m [A_m J_m(kr) + B_m Y_m(kr)] e^{im\theta} \quad (1.5)$$

For

$$\vec{A} = \frac{\Phi}{2\pi r} \hat{\theta} \Rightarrow S(\vec{x}) = \frac{q\Phi}{2\pi} \theta \equiv \alpha\theta \quad (1.6)$$

and single-value condition, then

$$\begin{aligned} e^{i\alpha(\theta+2\pi)} \varphi(r, \theta + 2\pi) &= e^{i\alpha\theta} \varphi(r, \theta) \\ \Rightarrow \varphi(r, \theta + 2\pi) &= e^{-2\pi i\alpha} \varphi(r, \theta) \\ \Rightarrow m + \alpha &= m' \in \mathbb{Z} \end{aligned} \quad (1.7)$$

Then

$$\varphi(r, \theta) = \sum_{m=-\infty}^{\infty} A_m J_{|m-\alpha|}(kr) e^{i(m-\alpha)\theta} \quad (1.8)$$

The Neumann function is excluded, for $Y(0) = \infty$ and the radius of magnetic flux tube is zero.

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When $r \rightarrow \infty$

$$\varphi \rightarrow \varphi^A = \sum_{m=-\infty}^{\infty} A_m \sqrt{\frac{2}{\pi k r}} \cos\left(kr - \frac{|m - \alpha|\pi}{2} - \frac{\pi}{4}\right) e^{i(m-\alpha)\theta} \quad (2.1)$$

And 2-dimensional scattering wave function must be like

$$\varphi^A = e^{ikr \cos \theta} e^{-i\alpha\theta} + \frac{f(\theta)}{\sqrt{r}} e^{ikr} \quad (2.2)$$

Then

$$\begin{aligned} e^{ikr \cos \theta} &= \sum_{m=-\infty}^{\infty} i^m J_m(kr) e^{im\theta} \\ &\stackrel{r \rightarrow \infty}{\sim} \sum_{m=-\infty}^{\infty} i^m \sqrt{\frac{2}{\pi k r}} \cos\left(kr - \frac{|m|\pi}{2} - \frac{\pi}{4}\right) e^{im\theta} \end{aligned} \quad (2.3)$$

Note that e^{-ikr} term is only contributed by planar wave term. Then

$$A_m = i^m e^{i\delta_m} \quad (2.4)$$

where $\delta_m = \frac{\pi}{2} (|m| - |m - \alpha|)$. Then $f(\theta)$ can be caculated by matching e^{ikr} term.

$$\begin{aligned} f(\theta) &= \sum_{m=-\infty}^{\infty} i^m \frac{1}{\sqrt{2\pi k}} \left(e^{2i\delta_m} - 1 \right) e^{-i\left(\frac{|m|\pi}{2} + \frac{\pi}{4}\right)} e^{i(m-\alpha)\theta} \\ &= \sum_{m=-\infty}^{\infty} \frac{1}{\sqrt{2\pi k}} \left(e^{2i\delta_m} - 1 \right) e^{-i\left(\frac{|m|-m}{2}\pi + \frac{\pi}{4}\right)} e^{i(m-\alpha)\theta} \end{aligned} \quad (2.5)$$

Assuming $0 < \alpha < 1$,

$$\sqrt{2\pi k} e^{i\frac{\pi}{4}} e^{i\alpha\theta} f(\theta) \equiv \sum_{m=-\infty}^{\infty} g(m) \quad (2.6)$$

where

$$g(m) = \begin{cases} (e^{i\alpha\pi} - 1) e^{im\theta} & m \geq \alpha \\ (e^{i\pi(2m-\alpha)} - 1) e^{im\theta} & 0 \leq m < \alpha \\ (e^{-i\alpha\pi} - 1) e^{im\pi} e^{im\theta} & m < 0 \end{cases} \quad (2.7)$$

Poisson summation:

$$\begin{aligned} \sum_{m=-\infty}^{\infty} g(m) &= \sum_{n=-\infty}^{\infty} \int g(\mu) e^{-2\pi n i \mu} d\mu \\ &= \sum_{n=-\infty}^{\infty} \{ (e^{i\alpha\pi} - 1) e^{i\alpha(\theta-2\pi n)} \pi \delta(\theta - 2\pi n) \\ &\quad + \frac{1}{i(2\pi(1-n) + \theta)} \left((e^{i\alpha\pi} - 1) e^{i\alpha(\theta-2\pi n)} - e^{-i\alpha\pi} + 1 \right) \\ &\quad + (e^{-i\alpha\pi} - 1) \pi \delta(\theta + \pi - 2\pi n) \} \\ &\stackrel{\theta \neq k\pi}{=} \sum_{n=-\infty}^{\infty} \frac{1}{i(2\pi(1-n) + \theta)} \left((e^{i\alpha\pi} - 1) e^{i\alpha(\theta-2\pi n)} - e^{-i\alpha\pi} + 1 \right) \end{aligned} \quad (2.8)$$

Attention that the residues of

$$h(z) = \frac{1}{i(2\pi(1-z) + \theta)} \left((e^{i\alpha\pi} - 1) e^{i\alpha(\theta-2\pi z)} - e^{-i\alpha\pi} + 1 \right) \pi \cot(\pi z) \quad (2.9)$$

are

$$\begin{aligned} \text{Res}_{z=n} h(z) &= \frac{1}{i(2\pi(1-n) + \theta)} \left((e^{i\alpha\pi} - 1) e^{i\alpha(\theta-2\pi n)} - e^{-i\alpha\pi} + 1 \right) \\ \text{Res}_{z=1+\frac{\theta}{2\pi}} &= -\frac{1}{2i} (1 - e^{-2\pi i\alpha}) \cot\left(\frac{1}{2}\theta\right), \end{aligned} \quad (2.10)$$

then

$$\sum_{m=-\infty}^{\infty} g(m) = \frac{1}{2i} (1 - e^{-2\pi i\alpha}) \cot\left(\frac{1}{2}\theta\right) \quad (2.11)$$